DOCKER DAY 6 – TOMCAT ASSIGNMENT

Custom index.html • Apache Tomcat • Port Mapping • Docker Hub

Objective

Build a custom Docker image from the official Apache Tomcat image, serve a simple index.html page, run multiple containers with different host ports, verify browser access, and push the image to Docker Hub. No WAR file, Maven, or Nginx is required.

Prerequisites

- An EC2 instance with Docker installed and running (or another Docker host).
- SSH access to the EC2 instance.
- Inbound TCP ports 8081–8083 allowed in the EC2 security group, preferably restricted to your own IP.
- A Docker Hub account for the optional push/pull section.

```
[root@ip-172-31-5-170 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-5-170 ~]# docker images
REPOSITORY   TAG          IMAGE ID       CREATED      SIZE
tomcat       10.1-jdk17   e0014f6cd9d4   6 days ago   478MB
[root@ip-172-31-5-170 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[root@ip-172-31-5-170 ~]#
```

Task 1 — Create the project and HTML page

Run these commands on your Docker host:

```
mkdir -p ~/docker-tomcat
cd ~/docker-tomcat
```

Create a simple web page:

```
cat > index.html <<'EOF'
<!DOCTYPE html>
<html>
<head><title>Docker Day 6</title></head>
<body style="font-family: Arial; text-align: center;">
  <h1>Welcome to Docker Day 6!</h1>
  <h2>Apache Tomcat Container</h2>
  <p>My custom HTML page is running inside Docker.</p>
</body>
</html>
EOF
```

Verify the file:

```
cat index.html
```

SCREENSHOT PLACEHOLDER — index.html content

```
[root@ip-172-31-5-170 docker-tomcat]# pwd
/root/docker-tomcat
[root@ip-172-31-5-170 docker-tomcat]# cat > index.html <<'EOF'
<!DOCTYPE html>
<html>
<head><title>Docker Day 6</title></head>
<body style="font-family: Arial; text-align: center;">
  <h1>Welcome to Docker Day 6!</h1>
  <h2>Apache Tomcat Container</h2>
  <p>My custom HTML page is running inside Docker.</p>
</body>
</html>
EOF
[root@ip-172-31-5-170 docker-tomcat]# cat index.html
<!DOCTYPE html>
<html>
<head><title>Docker Day 6</title></head>
<body style="font-family: Arial; text-align: center;">
  <h1>Welcome to Docker Day 6!</h1>
  <h2>Apache Tomcat Container</h2>
  <p>My custom HTML page is running inside Docker.</p>
</body>
</html>
[root@ip-172-31-5-170 docker-tomcat]#
```

Paste your screenshot above. Suggested evidence: index.html content.

Task 2 — Create the Dockerfile

This Dockerfile uses Tomcat to serve index.html at the root URL. It does not require a WAR file.

```
cat > Dockerfile <<'EOF'
FROM tomcat:10.1-jdk17
RUN rm -rf /usr/local/tomcat/webapps/*
RUN mkdir -p /usr/local/tomcat/webapps/ROOT
COPY index.html /usr/local/tomcat/webapps/ROOT/index.html
EXPOSE 8080
EOF
```

Review the Dockerfile:

```
cat Dockerfile
```

SCREENSHOT PLACEHOLDER — Dockerfile

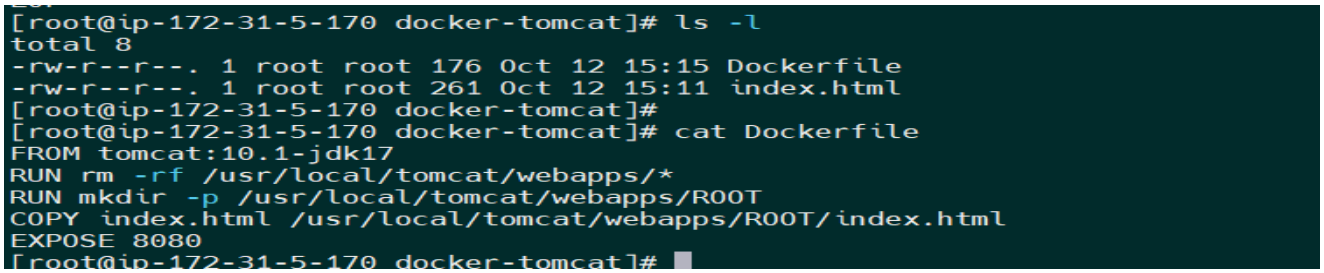
te your screenshot above. Suggested evidence: Dockerfile.

Task 3 — Build a custom image

```
docker build -t my-tomcat:v1 .
docker images
```

Expected result: the image my-tomcat with tag v1 appears in the image list.

SCREENSHOT PLACEHOLDER — Successful docker build and docker images output



Paste your screenshot above. Suggested evidence: Successful docker build and docker images output.

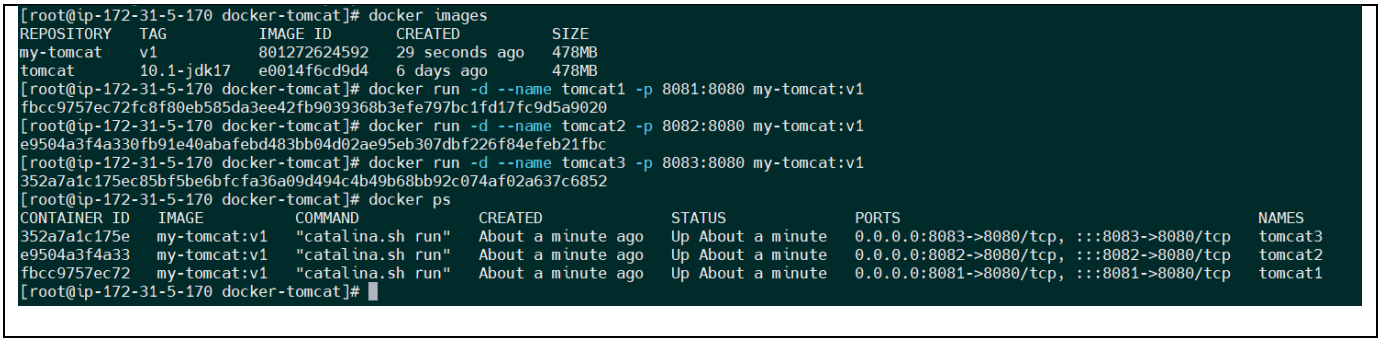
Task 4 — Run three containers with port mapping

```
docker run -d --name tomcat1 -p 8081:8080 my-tomcat:v1
docker run -d --name tomcat2 -p 8082:8080 my-tomcat:v1
docker run -d --name tomcat3 -p 8083:8080 my-tomcat:v1
docker ps
```

Port mapping means HOST_PORT:CONTAINER_PORT. Each container listens on Tomcat port 8080 internally; the host exposes ports 8081, 8082, and 8083.

Container	Host port	Container port
tomcat1	8081	8080
tomcat2	8082	8080
tomcat3	8083	8080

SCREENSHOT PLACEHOLDER — Three running containers and port mappings



Paste your screenshot above. Suggested evidence: Three running containers and port mappings.

Task 5 — Test the application

From the EC2 host, test locally:

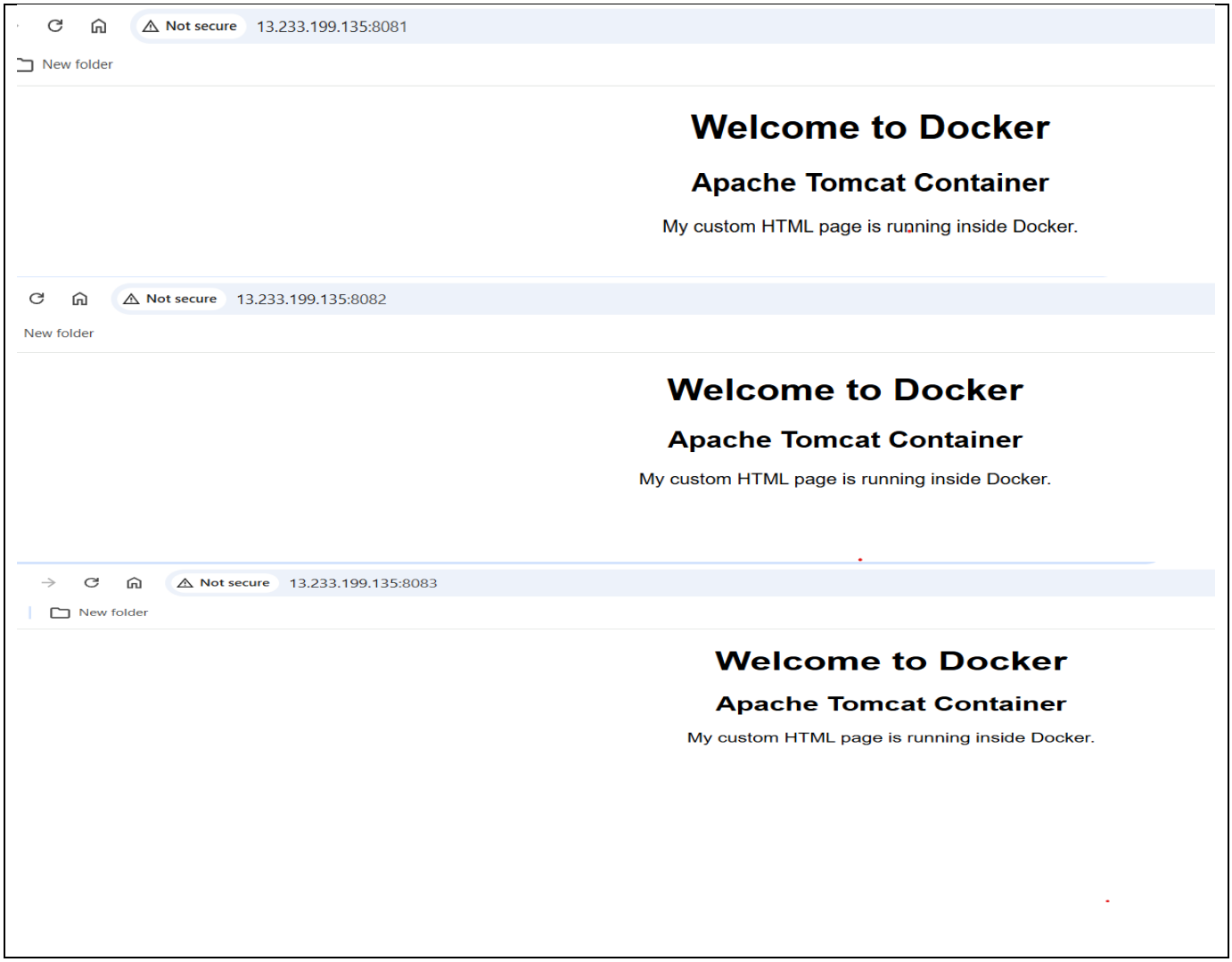
```
curl http://localhost:8081
curl http://localhost:8082
curl http://localhost:8083
```

From your browser, replace EC2_PUBLIC_IP with your instance public IPv4 address:

```
http://EC2_PUBLIC_IP:8081
http://EC2_PUBLIC_IP:8082
http://EC2_PUBLIC_IP:8083
```

If it does not load, confirm the EC2 security group allows inbound TCP 8081–8083 from your IP and check logs with docker logs tomcat1.

SCREENSHOT PLACEHOLDER — Browser showing Welcome to Docker Day 6



Paste your screenshot above. Suggested evidence: Browser showing Welcome to Docker Day 6.